

The Number of Points in Marcinkiewicz-Zygmund Inequalities

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L_2 MZ inequalities [5]

Let $\bullet$ Ω be domain with measure μ ,

- $L_2 = L_2(\Omega, \mu)$ Lebesgue space, and
- V an m -dimensional function space.

Points $x_1, \dots, x_n$ and weights $\omega_1, \dots, \omega_n > 0$ are said to fulfill an **L_2 MZ inequality** with constants $0 < 1 - \varepsilon \leq 1 + \varepsilon$, iff for all $f \in V$

$$(1 - \varepsilon) \|f\|_{L_2}^2 \leq \sum_{i=1}^n \omega_i |f(x_i)|^2 \leq (1 + \varepsilon) \|f\|_{L_2}^2. \quad (\text{MZ})$$

Two examples on $\mathbb{T}^d$

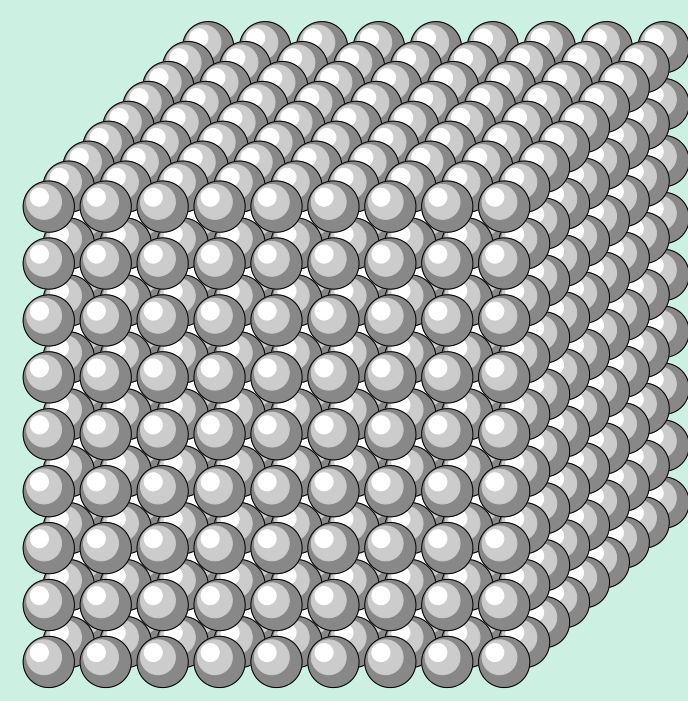
Equispaced point grid

Let $\bullet$ $L_2 = L_2(\mathbb{T}^d, d\mathbf{x})$, $\ell \in 2\mathbb{N}$, $n = \ell^d$

- $I = \left\{ -\frac{\ell}{2}, \dots, \frac{\ell}{2} - 1 \right\}^d$ cube of frequencies,
- $V = \text{span}\{\exp(2\pi i \langle \mathbf{k}, \cdot \rangle)\}_{\mathbf{k} \in I}$.

Then for $n = |I|$ (MZ) holds with $\varepsilon = 0$ and $\omega_i = \frac{1}{n}$ for

$$\mathbf{X} = \left\{ \frac{1}{\ell} \mathbf{i} : \mathbf{i} \in \{1, \dots, \ell\}^d \right\}$$



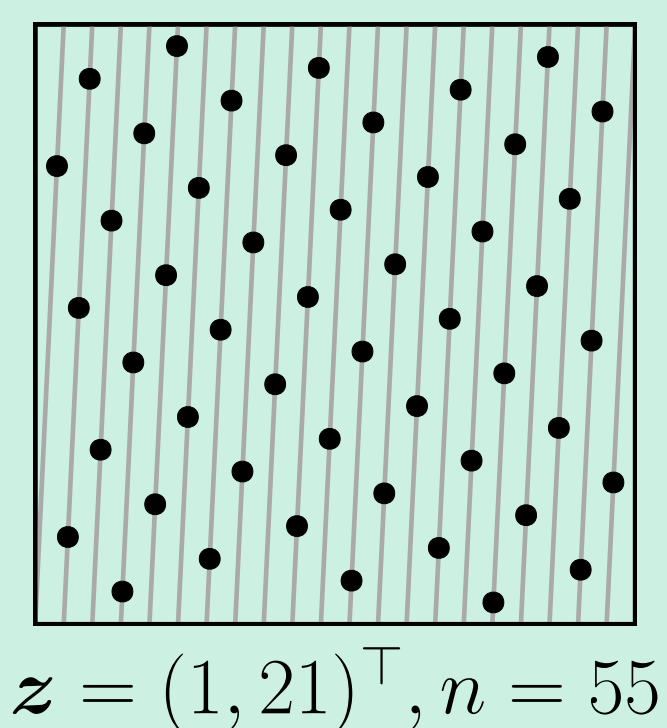
Rank-1 lattice

Let $\bullet$ $L_2 = L_2(\mathbb{T}^d, d\mathbf{x})$,

- $I \subseteq [-|I|/2, |I|/2]^d \subseteq \mathbb{Z}^d$, symmetric
- $V = \text{span}\{\exp(2\pi i \langle \mathbf{k}, \cdot \rangle)\}_{\mathbf{k} \in I}$.

Then for some $|I| \leq n \leq |I|^2$ (MZ) holds with $\varepsilon = 0$ and $\omega_i = \frac{1}{n}$ for the rank-1 lattice

$$\mathbf{X} = \left\{ \frac{i}{n} \mathbf{z} \bmod \mathbf{1}, i = 1, \dots, n \right\}$$



Namesake

Józef Marcinkiewicz (1910–1940)

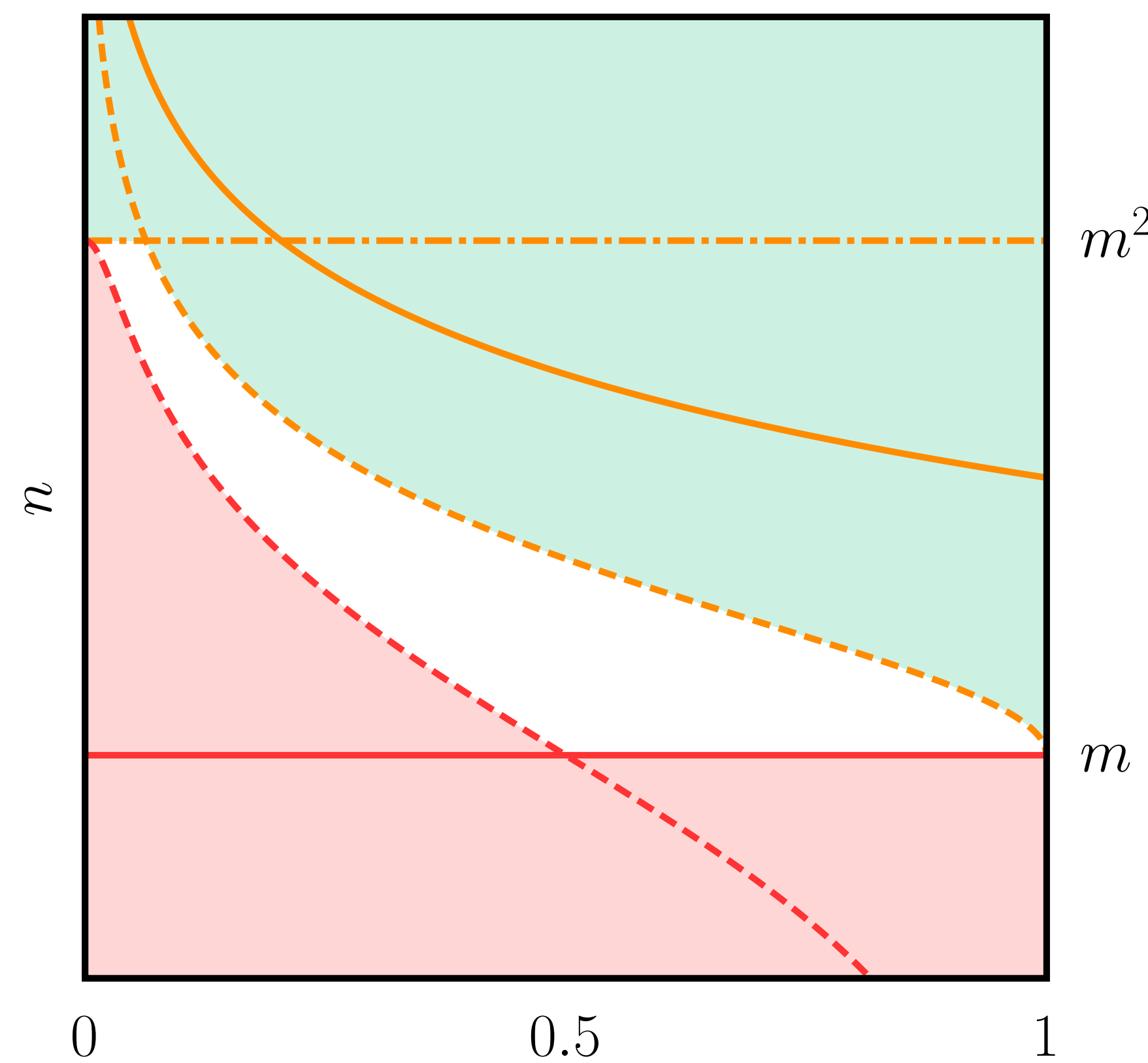


- born in Cimoszka, Poland
- student of Zygmund
- joint army during German invasion
- taken as POW during Soviet invasion
- believed to be murdered by the NKVD in Kharkiv

Antoni Zygmund (1900–1992)



- born in Warsaw, Poland
- emigrated to the US in 1940
- until 1945 Mount Holyoke College
- until 1947 University of Pennsylvania
- finally at University of Chicago



- Christoffel sampling
- BSS subsampling
- Carathéodory subsampling
- dimension argument
- ETF construction

Connection to frames

- Let $\bullet$ $\{\eta_1, \dots, \eta_m\}$ be ONB of V ,
- $\mathbf{y}^i = (\eta_k(x_i))_{k=1}^m$, and
 - $f = \sum_{i=1}^m a_i \eta_k$.

By Parseval's identity, (MZ) becomes

$$(1 - \varepsilon) \|\mathbf{a}\|_2^2 \leq \sum_{i=1}^n \omega_i |\langle \mathbf{a}, \mathbf{y}^i \rangle|^2 \leq (1 + \varepsilon) \|\mathbf{a}\|_2^2$$

and $\forall f \in V$ becomes $\forall \mathbf{a} \in \mathbb{C}^m$.

Upper bounds

Christoffel sampling [3]

- Let $\bullet$ $\{\eta_1, \dots, \eta_m\}$ be ONB of V ,
- x_i i.i.d. w.r.t. $\frac{1}{m} \sum_{k=1}^m |\eta_k(x)|^2 d\mu$, and
 - $\omega_i = m(\sum_{k=1}^m |\eta_k(x_i)|^2)^{-1}$.

For $\varepsilon > 0$ (MZ) holds with

$$n \geq \frac{6m \log m}{\varepsilon^2}.$$

Tools: matrix concentration inequalities

BSS subsampling [1]

The BSS algorithm yields points and weights s.t. for $\varepsilon > 0$ (MZ) holds with

$$n = \left\lceil m \frac{1 + \sqrt{1 - \varepsilon^2}}{1 - \sqrt{1 - \varepsilon^2}} \right\rceil \leq \left\lceil \frac{4m}{\varepsilon^2} \right\rceil.$$

Tools: incremental picking of points and weights whilst controlling the frame bounds

Carathéodory subsampling [2, 6]

There exist points and weights s.t. (MZ) holds with $\varepsilon = 0$ and

$$n = \dim(\text{span}\{f \cdot \bar{g} : f, g \in V\}) \leq m^2.$$

Tools: transferring to $m \times m$ Hermitian matrices to use Carathéodory's theorem on convex cones

Lower bounds

Dimension argument

When $n < m$, there exists $f \in V$ s.t. $\|f\|_{L_2} \neq 0$ but $f(x_1) = \dots = f(x_n) = 0$. As (MZ) requires $1 - \varepsilon > 0$, this gives the lower bound

$$n \geq m.$$

ETF construction [WIP]

- Let $\bullet$ $\{\mathbf{y}^i\}_{i=1}^{m^2-m+1} \subseteq \mathbb{C}^m$ be as in (PG),
- $\Omega = \{1, \dots, m^2 - m + 1\}$,
 - μ the counting measure, and
 - $V = \text{span}\{\eta_k\}_{k=1}^m$ with

$$\eta_k: \Omega \rightarrow \mathbb{C}, i \mapsto (\mathbf{y}^i)_k.$$

For points $x_1, \dots, x_n$ and weights $\omega_1, \dots, \omega_n > 0$ fulfilling (MZ), it holds

$$n \geq \frac{m^2 - m + 1}{(\varepsilon/(1 - \varepsilon))^2 m + 1}.$$

The construction

Singer difference sets

- Let $\bullet$ $m - 1$ be a prime power,
- $N = m^2 - m + 1$.

1938 James Singer came up with a construction for $(N, m, 1)$ **difference sets**, i.e., $D \subseteq \mathbb{Z}/N\mathbb{Z}$ with $|D| = m$ such that

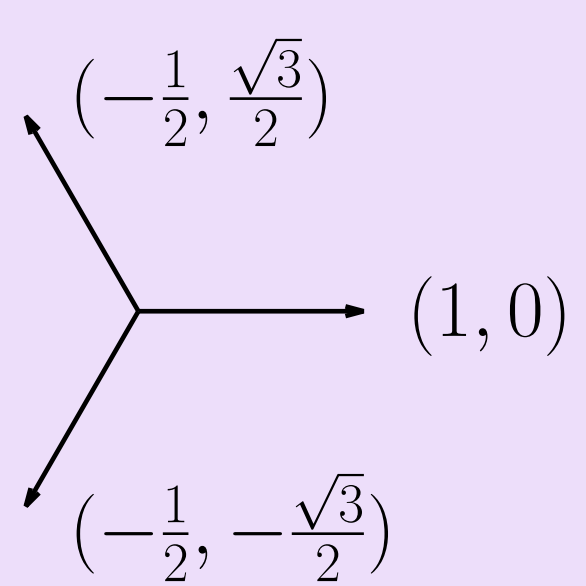
$$\{i - j : i \neq j \in D\} = \{\mathbb{Z}/N\mathbb{Z}\} \setminus \{0\}.$$

E.g. $\{0, 2, 12, 15, 16\}$ for $m = 5$ and $N = 21$.

Equiangular tight frames (ETFs)

A frame $\{\mathbf{y}^i\}_{i=1}^N \subseteq \mathbb{K}^N$ is called **equiangular**, iff

$$|\langle \mathbf{y}^i, \mathbf{y}^j \rangle| = \text{const} \quad i \neq j.$$



Using a Singer difference set, an ETF construction is given by

$$\mathbf{y}^j = \frac{1}{\sqrt{m}} \left(\exp(2\pi i j k / N) \right)_{k \in D} \in \mathbb{C}^m \quad (\text{PG})$$

for $j \in \{1, \dots, N = m^2 - m + 1\}$.

Welch bound

For a unit-norm frame, the following is true

$$\mu := \max_{i \neq j} |\langle \mathbf{y}^i, \mathbf{y}^j \rangle| \geq \sqrt{\frac{n - m}{m(n - 1)}}.$$

Equality only holds for ETFs and implies $n \leq m^2$.

Zauner's conjecture

- The existence of maximal ($n = m^2$) ETFs was conjectured 1999 by Gerhard Zauner.
- Constructions known for 115 values of m .

[1] F. Bartel, M. Schäfer, and T. Ullrich. Constructive subsampling of finite frames with applications in optimal function recovery. *Appl. Comput. Harmon. Anal.*, 2023.
[2] F. Bartel, L. Kämmerer, K. Pozharska, M. Schäfer, and T. Ullrich. Exact discretization, tight frames and recovery via D-optimal designs. *SMAI J. Comput. Math.*, 2025.
[3] A. Cohen, M.A. Davenport, and D. Leviatan. On the Stability and Accuracy of Least Squares Approximations. *Found. Comput. Math.*, 2017.

[4] L. Kämmerer, D. Potts, and T. Volkmer. Approximation of multivariate periodic functions by trigonometric polynomials based on rank-1 lattice sampling. *J. Complexity*, 2015.
[5] J. Marcinkiewicz and A. Zygmund. Sur les fonctions indépendantes. *Fund. Math.*, 1937.
[6] M. Schäfer and T. Ullrich. Beyond Tchakaloff Quadrature: Positive Functionals, Frames and Widths. *arXiv preprint*, 2025.